

Hemesh Raaja Malathi

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EDUCATION

Indiana University Bloomington

Masters in Computer Science: 3.84/4

Amrita Vishwa Vidyapeetham

Bachelors in Computer Science: 8.63/10

Bloomington, IN

August 2023 - May 2025

Chennai, India

August 2019 - May 2023

RELEVANT COURSEWORK

Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management System, Computer Architecture, Software Engineering, Distributed Systems, Cloud Computing

WORK EXPERIENCE

Software Engineering Intern

June 2024 - December 2024

Reinsurance Group of America

Chesterfield, Missouri

- Migrated outdated JavaScript codebase to React 18 with TypeScript and Material UI, enhancing maintainability and performance; updated tests with Mocha & React Testing Library, put into production via Jenkins
- Optimized Node.js backend with caching and efficient API design, boosting platform speed by 15%
- Automated UI testing for 2FA login, settings, and batch file submission in ETL platform using TypeScript, Cucumber JS, Playwright, and Axios API; incorporated into regression testing, cutting Human effort by 80%
- Fixed security vulnerabilities (SSDLC) while modernizing the frontend for compliance
- Designed and deployed an AWS Lambda function to automatically detect and terminate idle EC2 instances running EMR clusters, reducing cloud costs by \$1,000 per month; integrated with Datadog for real-time monitoring and Slack for automated notifications

Software Engineering Intern

February 2023 - April 2023

New Pro Data

Madurai, India

- Implemented an NLP-powered resume parsing system, achieving 95% accuracy in extracting key details like work experience, projects, education, and skills, reducing manual effort by 60%
- Implemented MPNet embeddings and FAISS for semantic matching between job descriptions and resume keywords, improving system accuracy to 93%
- Built a Django HR chatbot to automate onboarding, eliminating Workday and saving \$5,000–\$20,000 annually

Data Analyst Intern

August 2021 - September 2021

The Sparks Foundation

Chennai, India

- Optimized inventory using EDA, reducing costs by 20% and identifying key success factors.
- Identified pricing opportunities by analyzing sales data, addressing top three margin losses per category

PROJECTS

Cookbook: AI Recipe Generator | *React, Firebase, GroqCloud API, LLM(Llama Model)*

March 2025

- Developed a web application that allows users to search for recipes or generate personalized meal ideas using AI, leveraging GroqCloud's LLM(Llama model) for accurate recommendations.
- Enabled users to customize results based on cuisine, nutrition goals, and allergies, storing search history in Firebase for seamless access to previously generated recipes.

Movie Ticket Booking Application | *MERN Stack, Stripe*

January 2024 - May 2024

- Developed a full-stack movie ticket booking app with MERN stack, featuring separate User, Admin, and Theatre Owner interfaces, and JWT authentication with BCrypt hashing
- Integrated Ant Design for UI, Redux Toolkit for state management, and engineered real-time seat availability, show management, and Stripe payment processing with negative scenario handling
- Deployed on Render with GitHub integration, enabling seamless updates while providing an intuitive admin panel for user, ticket, and theatre management

TECHNICAL SKILLS

Languages: Python, Java, C/C++, R, TypeScript, JavaScript, HTML/CSS, PySpark

Frameworks & Databases: Flask, Django, Express.js, React, SQL, MongoDB, Neo4j, PostgreSQL

Tools: Git, Docker, Power BI, Tableau, Jenkins, Terraform, Kafka, Airflow, Hadoop, Kubernetes, dbt

Cloud: AWS (EC2, Dynamodb, S3, Athena, Redshift, Lambda, Glue), Azure, Databricks, Snowflake

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, RAG